```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

uploaded = files.upload()

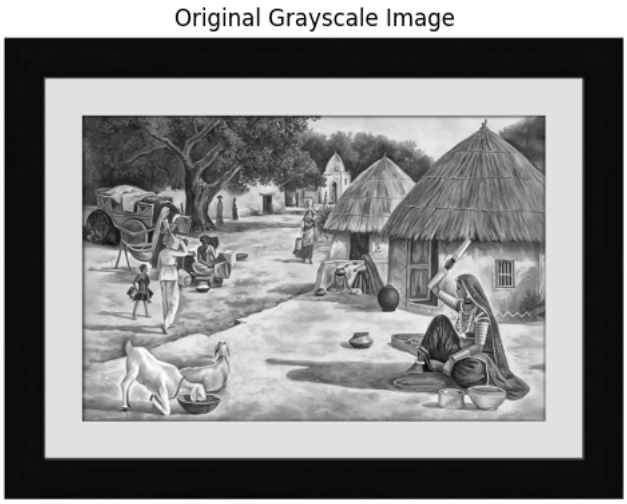
filename = next(iter(uploaded))
data = np.frombuffer(uploaded[filename], np.uint8)
img = cv2.imdecode(data, cv2.IMREAD_GRAYSCALE)

if img is None:
    raise ValueError("Please upload a valid image file.")

print("Image loaded successfully!")
print("Image dimensions:", img.shape)
```

```
Choose files village.jpg
village.jpg(image/jpeg) - 83279 bytes, last modified: 13/10/2026 - 100% done
Saving village.jpg to village.jpg
Image loaded successfully!
Image dimensions: (667, 894)
```

```
plt.figure(figsize=(6, 5))
plt.imshow(img, cmap="gray")
plt.title("Original Grayscale Image")
plt.axis("off")
plt.show()
```




```
dft = cv2.dft(
    np.float32(img),
    flags=cv2.DFT_COMPLEX_OUTPUT
)

dft_shift = np.fft.fftshift(dft, axes=(0, 1))

magnitude = cv2.magnitude(
    dft_shift[:, :, 0],
    dft_shift[:, :, 1]
)

spectrum = 20 * np.log(magnitude + 1)

plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(img, cmap="gray")
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(spectrum, cmap="gray")
```

```
plt.title("Fourier Magnitude Spectrum")
plt.axis("off")

plt.tight_layout()
plt.show()

print("DFT computed successfully.")
```




```
DFT computed successfully.
```

```
rows, cols = img.shape
crow, ccol = rows // 2, cols // 2

Y, X = np.ogrid[:rows, :cols]
distance = np.sqrt((X - ccol)**2 + (Y - crow)**2)

cutoff = min(rows, cols) * 0.10
low_pass_mask = (distance <= cutoff).astype(np.float32)

low_pass_shifted = dft_shift * low_pass_mask[:, :, np.newaxis]

low_pass_dft = np.fft.ifftshift(
    low_pass_shifted, axes=(0, 1)
)

low_pass_complex = cv2.idft(low_pass_dft)
low_pass_img = cv2.magnitude(
    low_pass_complex[:, :, 0],
    low_pass_complex[:, :, 1]
)

low_pass_img = np.clip(low_pass_img, 0, 255).astype(np.uint8)

plt.figure(figsize=(12, 4))

plt.subplot(1, 3, 1)
plt.imshow(low_pass_mask, cmap="gray")
plt.title("Low-Pass Mask")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(img, cmap="gray")
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.imshow(low_pass_img, cmap="gray")
plt.title("Low-Pass Filtered")
plt.axis("off")

plt.tight_layout()
plt.show()
```




```
high_pass_mask = 1.0 - low_pass_mask

high_pass_shifted = dft_shift * high_pass_mask[:, :, np.newaxis]

high_pass_dft = np.fft.ifftshift(
    high_pass_shifted, axes=(0, 1)
)

high_pass_complex = cv2.idft(high_pass_dft)
high_pass_img = cv2.magnitude(
    high_pass_complex[:, :, 0],
    high_pass_complex[:, :, 1]
)

high_pass_img = cv2.normalize(
    high_pass_img, None, 0, 255, cv2.NORM_MINMAX
).astype(np.uint8)

plt.figure(figsize=(12, 4))

plt.subplot(1, 3, 1)
plt.imshow(high_pass_mask, cmap="gray")
plt.title("High-Pass Mask")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(img, cmap="gray")
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.imshow(high_pass_img, cmap="gray")
plt.title("High-Pass Filtered")
plt.axis("off")

plt.tight_layout()
plt.show()
```




```
plt.figure(figsize=(14, 9))

plt.subplot(2, 2, 1)
```

```
plt.imshow(img, cmap="gray")
plt.title("Original Image")
plt.axis("off")

plt.subplot(2, 2, 2)
plt.imshow(spectrum, cmap="gray")
plt.title("Magnitude Spectrum")
plt.axis("off")

plt.subplot(2, 2, 3)
plt.imshow(low_pass_img, cmap="gray")
plt.title("Low-Pass Filtered Image")
plt.axis("off")

plt.subplot(2, 2, 4)
plt.imshow(high_pass_img, cmap="gray")
plt.title("High-Pass Filtered Image")
plt.axis("off")

plt.tight_layout()
plt.show()

print("Comparison completed successfully.")
```




```
Comparison completed successfully.
```

```
cv2.imwrite("original_image.png", img)
cv2.imwrite("low_pass_result.png", low_pass_img)
cv2.imwrite("high_pass_result.png", high_pass_img)

print("Output images saved successfully.")

files.download("low_pass_result.png")
```

```
files.download("high_pass_result.png")
```

```
Output images saved successfully.
```

```
qa = [
("Q1. What is the Fourier Transform? Why is it important in digital image processing?",
"The Fourier Transform converts an image from the spatial domain into its frequency components. It helps analyze image p

("Q2. Differentiate between the spatial domain and the frequency domain.",
"The spatial domain represents an image using pixel intensities and processes pixels directly. The frequency domain repr

("Q3. What is the significance of the Discrete Fourier Transform (DFT) in image processing?",
"The DFT converts a digital image into a frequency representation. It helps analyze frequency components and implement f

("Q4. Explain the purpose of shifting the zero-frequency component to the center.",
"Shifting the zero-frequency component to the center makes the frequency spectrum easier to visualize. Low frequencies a

("Q5. Compare Low-Pass and High-Pass Frequency Filters with applications.",
"Low-Pass Filters retain low frequencies and suppress high frequencies. They are useful for smoothing and noise reductio

("Q6. What is the role of the Inverse Fourier Transform (IDFT) in image reconstruction?",
"The Inverse Discrete Fourier Transform converts a filtered frequency representation back into the spatial domain, produ

("Q7. Why is frequency domain filtering preferred for certain image enhancement tasks?",
"It allows selective manipulation of frequency components. It is useful for suppressing periodic noise, smoothing images

("Q8. Mention four real-world applications of Fourier Transform in computer vision and image analysis.",
"1. Medical image enhancement and restoration.\n2. Satellite and remote sensing image analysis.\n3. Image denoising and

("Q9. Compare frequency domain filtering with spatial domain filtering.",
"Spatial filtering directly modifies pixels using neighborhoods and kernels. Frequency domain filtering transforms the i

("Q10. How does frequency domain filtering improve image restoration and feature extraction?",
"It can suppress unwanted frequency components and emphasize useful image structures. This may reduce noise, improve edg
]

print("EXPERIMENT NO. 4")
print("FREQUENCY DOMAIN IMAGE FILTERING USING FOURIER TRANSFORM")
print("=" * 70)

for question, answer in qa:
    print("\n" + question)
    print("-" * 70)
    print(answer)
    print()

print("All 10 questions and answers completed successfully.")
```

```
EXPERIMENT NO. 4
FREQUENCY DOMAIN IMAGE FILTERING USING FOURIER TRANSFORM
======================================================================

Q1. What is the Fourier Transform? Why is it important in digital image processing?
----------------------------------------------------------------------
The Fourier Transform converts an image from the spatial domain into its frequency components. It helps analyze image


Q2. Differentiate between the spatial domain and the frequency domain.
----------------------------------------------------------------------
The spatial domain represents an image using pixel intensities and processes pixels directly. The frequency domain rep


Q3. What is the significance of the Discrete Fourier Transform (DFT) in image processing?
----------------------------------------------------------------------
The DFT converts a digital image into a frequency representation. It helps analyze frequency components and implement


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----------------------------------------------------------------------
Shifting the zero-frequency component to the center makes the frequency spectrum easier to visualize. Low frequencies


Q5. Compare Low-Pass and High-Pass Frequency Filters with applications.
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Low-Pass Filters retain low frequencies and suppress high frequencies. They are useful for smoothing and noise reducti


Q6. What is the role of the Inverse Fourier Transform (IDFT) in image reconstruction?
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The Inverse Discrete Fourier Transform converts a filtered frequency representation back into the spatial domain, pro
```

```
Q7. Why is frequency domain filtering preferred for certain image enhancement tasks?
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It allows selective manipulation of frequency components. It is useful for suppressing periodic noise, smoothing imag


Q8. Mention four real-world applications of Fourier Transform in computer vision and image analysis.
--------------------------------------------------------------------------
1. Medical image enhancement and restoration.
2. Satellite and remote sensing image analysis.
3. Image denoising and restoration.
4. Biometric image analysis.


Q9. Compare frequency domain filtering with spatial domain filtering.
--------------------------------------------------------------------------
Spatial filtering directly modifies pixels using neighborhoods and kernels. Frequency domain filtering transforms the


Q10. How does frequency domain filtering improve image restoration and feature extraction?
--------------------------------------------------------------------------
It can suppress unwanted frequency components and emphasize useful image structures. This may reduce noise, improve e

All 10 questions and answers completed successfully.
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